

Compressed Sampling for Spectrum Measurement and DOA Estimation With Array Cooperative MWC

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Abstract—Spectrum measurement and direction of arrival (DOA) estimation methods based on traditional Nyquist theory lead to high sampling rate and a large amount of data, which bring challenges to hardware and signal processing. An array cooperative modulated wideband converter (ACMWC) sub-Nyquist sampling system for spectrum measurement and DOA estimation is proposed in this work. Based on a linear array (LA) cooperative MWC sampling structure with an arbitrary array element position, we construct a unified space–frequency-domain sparse model by combining the modulation information of different sensors and the spatial phase information of the array to estimate carrier and DOA simultaneously. Combining multi-sensor data to jointly reconstruct spectrum and estimate DOA can improve data utilization and estimation accuracy. In addition, we also give a spatial separation processing scheme based on the sparsity characteristics of signal spatial distribution, which can greatly reduce the computational complexity. Compared with similar approaches, the problem of source ambiguity and coherent signals in array parameter estimation can be solved by the time–space distribution characteristics of spectrum and DOA. Moreover, the proposed method relaxes the restrictions on array manifolds and can be used for LA with arbitrary sensor positions with wider application prospects. The effectiveness of the proposed method is verified by simulation and hardware results.

Index Terms—Array cooperative modulated wideband converter (ACMWC), compressed sensing (CS), direction of arrival (DOA), spectrum measurement, sub-Nyquist sampling.

NOMENCLATURE

Symbol	
$(\cdot)^T$	Transpose.
$(\cdot)^H$	Conjugate transpose.
$\otimes$	Kronecker product.
$\odot$	Khatri–Rao product.
$(\cdot)^\dagger$	Pseudoinverse.
$\ \cdot\ _p$	p -norm.

$\lceil \cdot \rceil$	Ceiling.
$\lfloor \cdot \rfloor$	Floor.
$\text{mod}(\cdot)$	Modulo.
Acronyms	
CP	CANDECOMP/PARAFAC.
CS	Compressed sensing.
DOA	Direction of arrival.
LA	Linear array.
LPF	Low-pass filter.
MWC	Modulated wideband converter.
ACMWC	Array cooperative MWC.
SOMP	Simultaneous orthogonal matching pursuit.
ULA	Uniform LA.

I. INTRODUCTION

THE problem of spectrum measurement and DOA estimation is an important research topic for radar, sonar, cognitive radio, and other engineering measurement applications [1], [2], [3]. At present, many scholars have considered the joint estimation of frequency and DOA based on the Nyquist sampling theory, such as in [4], [5], [6], and [7]. The spectrum under monitoring is very wide in many modern applications. The sampling schemes based on the traditional Nyquist theory lead to high sampling rate and a large amount of data, which bring challenges to hardware and signal processing. To overcome the bottleneck of sampling rate, scholars have proposed many sub-Nyquist sampling structures based on CS [8], [9], such as multi-coset sampler [10] and MWC [11], [12]. Fang et al. [13] provide an overview of recent advances on sub-Nyquist sampling wideband spectrum sensing techniques and present an elaborate discussion of the pros and cons of each approach. These methods can sample and reconstruct the wideband signal at a sampling rate much lower than Nyquist rate.

The sub-Nyquist sampling method brings a new solution to joint estimation of frequency and DOA estimation for wideband signals. Some works have considered the joint estimation of frequency, DOA, and spectrum based on sub-Nyquist sampling. Different from the wideband spectrum sensing methods in [10], [11], [12], and [13], the signals received by each sensor in array are not identical as the target signals have different spatial delays depending on their DOA and frequency. A sim-

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ple sub-Nyquist sampling receiver was proposed in [14], which estimated the carrier frequency using the phase difference in the direct and delayed paths, and then estimated the DOA by the multiple signal classification (MUSIC) method. Fang et al. [15] and Wang et al. [16] proposed a phased array sub-Nyquist structure using flexible time delays. And a DOA, carrier, and power spectrum estimation on the basis of CP decomposition are developed. The architecture is simple and easy to implement, but performance decreases when the time difference between the antenna delays is too small. Liu et al. [17], [18] and Liu and Wei [19] proposed a carrier and DOA recovery method based on multi-coset sampler. And a novel time-space model is proposed to improve phase estimation performance. However, a large number of channels are required, resulting in high hardware cost. These time-delayed [14], [15], [16] or multi-coset sampler [17], [18], [19]-based methods face practical problems. The time delays require high accuracy, and the analog front-end of low rate ADCs must tolerate the input bandwidth [11], which is difficult to implement with current RF hardware.

The MWC theory was proposed to overcome these problems [20]. Ioushua et al. [20] and Stein et al. [21] applied MWC to DOA estimation and proposed a carrier, DOA estimation, and signal reconstruction method on an L-shaped array MWC system. The sensors consist of an analog mixing front-end and one MWC channel. They provide three reconstruction algorithms, based on CP decomposition, parameter estimation via the rotational invariance technique (ESPRIT) method, and CS technique, respectively. The system used an array manifold to estimate carrier and DOA parameters, but did not make full use of MWC in processing multiband signals to obtain frequency-domain information, resulting in low efficiency. In [22] and [23], a ULA MWC system is proposed for carrier and DOA estimation. A special phase relation is constructed to estimate carrier using a cycle-shifted pseudorandom sequences. However, this method is unable to reconstruct the signal and can only process one signal. A two-stage estimation algorithm based on array MWC has been proposed in [24]. The carriers are estimated from the recovered signal of an MWC structure, and the DOAs are estimated from the array signals with the MUSIC algorithm. It can solve the problem of source ambiguity, but the matching method is tedious, and the signal spectrum cannot be overlapped. Joshi et al. [2], [25] introduced on-line learning based on [24], to realize reconstruction without signal distribution prior knowledge. However, the channels of the array MWC are only used to estimate DOA with low utilization efficiency. We proposed a beamforming MWC system in [26] that exploits the spatial filtering and signal enhancement properties of adaptive beamforming, which enhances the sparsity property of the signals and greatly simplifies the MWC structure. However, it cannot be applied to complex scenarios where coherent and ambiguous sources are present. Zhang et al. [27] proposed a joint spectrum measurement and DOA estimation algorithms based on extended channel LA MWC. The hardware structure is simple and can recognize more sources than the sensors, but it assumed that each source is considered to lie entirely in a subband, which is difficult to achieve in practical applications.

These two methods [26], [27] can make full use of the ability of the MWC structure to process sparse signals in the frequency domain, and the structures are simple and effective.

However, almost all the existing methods assume the absence of source ambiguity and coherent signals. The source ambiguity is caused by the same electronic angles from different sources leading to a linear correlation of the steering vectors [20], [24]. And coherent signals often exist due to multipath propagation, or enemy jamming [28], [29] in practice. These two problems can lead to rank deficiency of the array manifold matrix or cross correlation matrix, which invalidate the above sub-Nyquist sampling estimation method.

In this article, we propose a novel ACMWC system for spectrum measurement and DOA estimation. Different from the existing researches, we can combine data from multiple array elements to jointly estimate the carrier frequency and DOA simultaneously using the CS method based on an LA MWC sampling structure with different mixing sequences in each channel. Compared with the previous researches, the proposed system does not require a uniform array and can deal with the presence of source ambiguity and coherent signals, which relaxes the restrictions on the array manifolds and signal model, with wide application prospects. The contributions of this article are listed as follows.

- 1) An ACMWC sub-Nyquist sampling system is proposed for spectrum measurement and DOA estimation. By combining the modulation information of different sensors and the spatial phase information of the array to build a unified space-frequency-domain sparse model, DOA estimation and spectrum sensing can be performed simultaneously. Combining multisensor frequency and spatial data to jointly reconstruct spectrum and estimate DOA can improve data utilization and estimation accuracy. In addition, we give a spatial separation processing scheme based on the sparsity characteristics of signal spatial distribution, which can greatly reduce the computational complexity.
- 2) Compared with previous similar researches, the problem of source ambiguity and coherent signals in the array parameter estimation can be solved by the time-space distribution characteristics of spectrum and DOA. Moreover, the proposed method relaxes the restrictions on array manifolds and can be used for LA with arbitrary sensor positions, which has a wider range of applications compared with ULA and 2-D L-shaped array.
- 3) A hardware prototype is implemented to verify the effectiveness of the proposed ACMWC system, and the nonideal characteristics present in the hardware devices were analyzed. The experimental results show that the proposed ACMWC system has a better performance on carrier estimation and signal reconstruction than other four studied methods. Besides, ACMWC can successfully estimate the parameters in the presence of source ambiguity and coherent signals.

The description of the symbols and the main acronyms used in this article is shown in Nomenclature section. The rest of this article is organized as follows. Section II introduces the

signal model. Section III gives the details of the proposed array receiver architecture, spectrum measurement, and DOA estimation method. Sections IV and V illustrate the simulation results and experimental results, respectively. Finally, concluding summaries are given in Section VI.

II. PROBLEM FORMULATION

Consider that M far-field narrowband signal sources $s_i(t)$ spread over a wide spectral range with Nyquist sampling rate f_{nyq} and unknown carrier frequencies f_i , where $i = 1, \dots, M$. The frequency bandwidth of signals $\{s_i(t)\}_{i=1}^M$ is limited to $\mathcal{B} = [-(B/2), (B/2)]$.

The signals $\{s_i(t)e^{j2\pi f_i t}\}_{i=1}^M$ impinge on the LA with N receiving sensors from distinct DOAs $\theta_i \in (-90^\circ, 90^\circ)$. The positions of the N sensors are denoted as a vector $\mathbf{d} = [d_1, d_2, \dots, d_N]$ with $d_1 = 0$, where d_n is the distance of the n th sensor relative to the first sensor. It is worth noting that different from the previous works [2], [15], [16], [18], [22], [23], [24], [25], [26], we do not require the sensors to be uniformly distributed. And we have no requirements for signal correlation, i.e., signal sources can be correlated or even coherent, which often happens in practice due to multipath propagation. Also, we do not strictly restrict the absence of source ambiguity, that is, we do not require $f_i \sin \theta_i \neq f_j \sin \theta_j$ for any $i \neq j$ as other works did. In contrast, our assumptions are obviously less restrictive to the array manifold and signal model, and more consistent with the situation in practical applications.

Under the narrowband assumption [20], the signal $u_n(t)$ received from the n th sensor can be presented as

$$u_n(t) \approx \sum_{i=1}^M s_i(t) e^{j2\pi f_i (t + \frac{d_n}{c} \sin \theta_i)} \quad (1)$$

where $n = 1, \dots, N$.

For convenience, we define the electronic angles as

$$\tau_i = f_i \sin \theta_i \quad (2)$$

and $(d_n/c)\tau_i$ denotes the spatial delay at the n th sensor with respect to the reference sensor. Then the Fourier transform of $u_n(t)$ is given by

$$U_n(f) = \sum_{i=1}^M S_i(f - f_i) e^{j2\pi \frac{d_n}{c} \tau_i}. \quad (3)$$

Our objective is to design an ACMWC system that can successfully reconstruct the signal spectrum $S_i(f)$ and estimate the signal carriers f_i and DOAs θ_i . In addition, we desire the proposed system to remain effective in the presence of coherent signals and source ambiguity.

III. PROPOSED ACMWC SYSTEM

A. System Description and Signal Analysis

The proposed receiver consists of an LA with N sensors as depicted in Fig. 1. The received signal $u_n(t)$ is first multiplied by a mixing sequence $p_n(t)$ with the periodicity of $T_p = 1/f_p$, and f_p is the periodic rate. Then, the signal is filtered by an LPF with cutoff frequency $1/(2T_s)$ and the filtered signal is

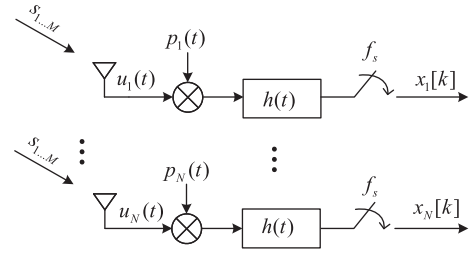


Fig. 1. LA MWC architecture with different mixing sequences in each receiving channel.

then sampled by ADC at a rate of $f_s = 1/T_s$. Define $\mathbf{x}[k] = [x_1[k], \dots, x_N[k]]^T$ as the samples with $k = 1, \dots, K$. For simplicity, $f_s = f_p$ is chosen.

Different from the existing works [2], [20], [21], [22], [23], [24], [25], [26], [27], we set up different mixing sequences in each receiving channel. Different mixing sequences allow us to fully exploit the mixing information of multiple channels to assist in signal supports' estimation, rather than using spatial information only to estimate the carrier frequency in previous works. Based on this feature, our system can deal with the situation of source ambiguity or coherent signals by combining mixing information and spatial information. Meanwhile, different mixing sequences increase the randomness of the sensor matrix, leading to the improvement of the robustness of the system. Therefore, the proposed architecture can improve the accuracy of carrier frequency estimation and improve the utilization efficiency of the MWC sub-Nyquist sampling structure by making full use of the MWC structure itself to deal with sparse signals in the frequency domain.

Consider the n th sensor. The Fourier expansion of mixing sequence $p_n(t)$ is given by

$$p_n(t) = \sum_{l=-\infty}^{\infty} c_{n,l} e^{j\frac{2\pi}{T_p} l t} \quad (4)$$

where $c_{n,l} = (1/T_p) \int_0^{T_p} p_n(t) e^{-j2\pi l f_p t} dt$ is the Fourier series coefficients of $p_n(t)$.

After mixing and low-pass filtering, the discrete-time Fourier transform (DTFT) of the n th sampling sequence $x_n[k]$ is

$$X_n(e^{j2\pi f T_s}) = \sum_{l=-L_0}^{L_0} c_{n,l} \sum_{i=1}^M S_i(f - f_i - l f_p) e^{j2\pi \frac{d_n}{c} \tau_i}, \quad f \in \mathcal{F}_s \quad (5)$$

where $\mathcal{F}_s = [-f_s/2, +f_s/2]$, $S_i(f)$ is the Fourier transform of $s_i(t)$, and $L_0 = \lceil (f_{\text{nyq}})/(2f_p) \rceil$ is the smallest integer that allow all nonzero elements of $X_n(f)$ contained in the summation.

Considering that the spectrum of a complex-valued signal contains only positive frequencies, that is, when $l = -L_0, \dots, -1$, $S_i(f - f_i - l f_p) = 0$. We only keep the part where $L \geq 0$ (5) can be simplified as

$$X_n(e^{j2\pi f T_s}) = \sum_{l=0}^{L_0} c_{n,l} \sum_{i=1}^M S_i(f - f_i - l f_p) e^{j2\pi \frac{d_n}{c} \tau_i}. \quad (6)$$

Define the spectral support set of target signals as $\mathbf{S} = [\mathbf{S}_1, \dots, \mathbf{S}_M]$, where $\mathbf{S}_i$ denotes the two adjacent indexes of the i th signal $s_i(t)$ over spectral continuous interval, with interval width f_p .

B. Signal Spectral Supports and Electronic Angles' Estimation

The theorem below gives a necessary condition for estimating spectral supports and electronic angles from the samples.

Theorem 1: Consider M far-field narrowband signals $\{s_i(t)\}_{i=1}^M$ impinging on an LA array with N sensors as in Fig. 1. If the number of sensors satisfies $N \geq 2M$, and the spectral supports $\mathbf{S}_i$, carrier frequencies f_i , and DOAs θ_i satisfy $\{\mathbf{S}_i, f_i \sin \theta_i\} \neq \{\mathbf{S}_j, f_j \sin \theta_j\}$. And the sampling rate of each channel $f_s \geq f_p \geq B$. Then the spectral supports of signals $\mathbf{S}$ and the electronic angles $\{\tau_i\}_{i=1}^M$ can be uniquely recovered from samples $\mathbf{x}[k]$.

Proof: Suppose that the electronic angles τ_i lie on a grid $\{\delta p\}_{p=1}^P$, where δ is a grid resolution parameter. Set the value range of δp to be the same as that of τ_i . Denote the signal $S_i(f - f_i)$ corresponding to $\tau_p = \delta p$ on the grid as $S_p(f)$. Then (6) can be further expressed as

$$\begin{aligned} X_n(e^{j2\pi f T_s}) &= \sum_{l=0}^{L_0} c_{n,l} \sum_{p=1}^P S_p(f - lf_p) e^{j2\pi \frac{d_n}{c} \delta p} \\ &= \sum_{l=0}^{L_0} c_{n,l} \sum_{p=1}^P e^{j2\pi \frac{d_n}{c} \delta p} \tilde{S}_{p,l}(f) \end{aligned} \quad (7)$$

where $\tilde{S}_{p,l}(f) = S_p(f - lf_p)$. When $\delta p = \tau_i$, we have

$$\tilde{S}_{p,l}(f) = S_i(f - f_i - lf_p). \quad (8)$$

Write (7) in matrix form as

$$X_n(e^{j2\pi f T_s}) = (\mathbf{c}_n \otimes \mathbf{a}_n) \tilde{\mathbf{S}}(f), \quad f \in \mathcal{F}_s \quad (9)$$

where $\otimes$ denotes the Kronecker product. $\mathbf{c}_n = [c_{n,0}, \dots, c_{n,L_0}]$, and $\mathbf{a}_n = [e^{j2\pi(d_n/c)\delta}, \dots, e^{j2\pi(d_n/c)\delta P}]$ is the steering vector of the n th sensor. $\tilde{\mathbf{S}}(f)$ is an $L_0 P \times 1$ unknown sparse vector, containing all L_0 spectrum slices corresponding to each grid point $p \in [1, \dots, P]$ of electronic angles. The elements in $\tilde{\mathbf{S}}(f)$ can be expressed as $\tilde{\mathbf{S}}(f) = [\tilde{S}_{1,0}(f), \dots, \tilde{S}_{P,0}(f), \dots, \tilde{S}_{1,L_0}(f), \dots, \tilde{S}_{P,L_0}(f)]^T$. Since $B \leq f_p$, the signal spectrum can be split by two adjacent bands with a bandwidth of f_p , and consequently $\tilde{\mathbf{S}}(f)$ has at most $2M$ nonzeros. So $\tilde{\mathbf{S}}(f)$ is $2M$ -jointly sparse vector over $f \in \mathcal{F}_s$.

Then, combining all the sensors can result in

$$\mathbf{X}(e^{j2\pi f T_s}) = (\mathbf{C}^T \odot \mathbf{A}^T)^T \tilde{\mathbf{S}}(f) = \mathbf{G} \tilde{\mathbf{S}}(f) \quad (10)$$

where $\odot$ is the Khatri-Rao product; $\mathbf{C}$ is an $N \times L_0$ matrix containing the Fourier series coefficients of mixing sequences $p_n(t)$, $n = 1, \dots, N$, with its n th row $\mathbf{c}_n$; $\mathbf{A}$ is an $N \times P$ array manifold matrix with its n th row $\mathbf{a}_n$; and $\mathbf{G} = (\mathbf{C}^T \odot \mathbf{A}^T)^T$ is an $N \times L_0 P$ matrix with $(n, l \cdot p)$ th element $G_{n,l \cdot p} = c_{n,l} e^{j2\pi(d_n/c)\delta p}$. The expansion of (10) is further given in (11), as shown at the bottom of the next page.

Equation (10) represents an infinite number of linear systems with joint sparsity, referred to as the infinite measurement

vectors (IMVs). Conceptually, its solution requires independent treatment of an infinite number of systems. We use the continuous to finite (CTF) block in [11] to transform the IMV problem into multiple measurement vectors (MMVs) problem for solving. Specifically, the first M eigenvalues of the correlation matrix $\mathbf{Q} = \mathbf{X}\mathbf{X}^H$ are multiplied by the corresponding eigenvectors to replace the samples, which is defined as a finite frame $\mathbf{v}$.

According to the CS theory, (10) has a unique solution if every $2M$ column of $\mathbf{G}$ is linearly independent. It can be observed that the lp th column of the perception matrix $\mathbf{G} = (\mathbf{C}^T \odot \mathbf{A}^T)^T$ can be expressed as the Hadamard product of the l th column of matrix $\mathbf{C}$ and the p th column of $\mathbf{A}$. With a random choice of signs, every $2M$ column of $\mathbf{C}$ is linearly independent with $N \geq 2M$ and $L > 2M$ [11], [30]. From (12), we can see that the array manifold matrix $\mathbf{A}$ is a Vandermonde matrix. So every $2M$ column of $\mathbf{G}$ is linearly independent as long as

$$c_{n,S_i} e^{j2\pi \frac{d_n}{c} f_i \sin \theta_i} \neq c_{n,S_j} e^{j2\pi \frac{d_n}{c} f_j \sin \theta_j} \quad (12)$$

where $\mathbf{S}_i = [S_{i,1}, S_{i,2}]$ and $\mathbf{S}_j = [S_{j,1}, S_{j,2}]$ denote the spectral supports of the i th and j th signals, respectively. The condition in (12) can be translated into for any $i \neq j$ have

$$\{\mathbf{S}_i, f_i \sin \theta_i\} \neq \{\mathbf{S}_j, f_j \sin \theta_j\}. \quad (13)$$

That is, as long as the spectral supports $\mathbf{S}_i$ or electronic angles $f_j \sin \theta_j$ of the signals are different, the signals can be successfully distinguished. In fact, the condition in (13) is easy to meet. Only if the carrier frequency and DOA of the two signals are almost the same, the spectral support and the electronic angle will be the same. In this case, these two signals can almost be considered as one signal. This can also be avoided by making the support sets different by choosing an appropriate spectral interval width f_p .

It can be seen that the proposed ACMWC system distinguishes the target signals by a combination of spectral support and electronic angle. Therefore, even if in the presence of source ambiguity (identical electronic angles for different signals) or coherent signals, the signals can be correctly estimated as long as the assumption in (13) is satisfied.

From the MWC theory [11], [30], the choice of $f_p \geq B$ ensures that each band can contribute only a single nonzero value to $\tilde{\mathbf{S}}(f)$. And to sample all the information of the signals in the baseband, it is necessary to satisfy $f_s \geq f_p$.

Define $\mathbf{Z} = [z_1, \dots, z_{2M}]$ as the index set of nonzero entries in $\tilde{\mathbf{S}}(f)$. We can obtain the support $\mathbf{Z}$ using the CS recovery algorithm such as SOMP method. It is worth noting that although $S_i(f)$ may be split into two adjacent bands, it is known from the signal model defined in (4) that a signal has only one electronic angle, which means that the spectral supports $S_{i,1}$ and $S_{i,2}$ correspond to the same one electronic angle τ_i . It is precisely because of this characteristic of the signal that we check the adjacent indices when looking for the support set. Once the support $\mathbf{Z}$ is determined, the corresponding spectral supports and the grid position indices

of electronic angles are given by

$$\hat{\mathbf{S}} = \text{mod}(\mathbf{Z}/L_0) - 1 \quad (14)$$

$$\hat{\mathbf{p}} = \lfloor \mathbf{Z}_{\text{odd}}/L_0 \rfloor + 1. \quad (15)$$

where $\mathbf{Z}_{\text{odd}}$ represents the odd-position elements of set $\mathbf{Z}$, and $\hat{\mathbf{S}} = [\hat{S}_1, \dots, \hat{S}_M]$ and $\hat{\mathbf{p}} = [\hat{p}_1, \dots, \hat{p}_M]$ are the estimated spectral supports and the grid position indices of electronic angles, respectively.

Then the electronic angles can be calculated as

$$\hat{\tau} = \left(\frac{\hat{\mathbf{p}}}{P} - 0.5 \right) f_{\text{nyq}} \quad (16)$$

where $\hat{\tau} = [\hat{\tau}_1, \dots, \hat{\tau}_M]$. ■

C. Spectrum Measurement, Carrier, and DOA Estimation

After $\mathbf{Z}$ is determined, the signals' spectrum is reconstructed by

$$\tilde{\mathbf{S}}_{\mathbf{Z}}(f) = \mathbf{G}_{\mathbf{Z}}^{\dagger} \mathbf{X}(f) \quad (17)$$

where $\tilde{\mathbf{S}}_{\mathbf{Z}}(f)$ contains only the entries of $\tilde{\mathbf{S}}$ indexed by $\mathbf{Z}$, and $\mathbf{G}_{\mathbf{Z}}^{\dagger}$ is the pseudoinverse of $\mathbf{G}_{\mathbf{Z}}$. Then according to (8), the spectrum of signal $s_i(t)$ can be expressed by

$$\hat{S}_i(f - f_i + S_i f_p) = \tilde{S}_{\hat{p}_i, S_i}(f). \quad (18)$$

The carrier frequency f_i is then estimated from the reconstructed spectrum $\hat{S}_i(f)$, $f \in [-f_p/2, +3f_p/2]$. First, we use the Welch method to obtain the signal power spectrum. Then, to improve the stability of spectrum estimation, a smoothing filtering was implemented on the Welch power spectrum $P_{s_i}(f)$ through taking average of U points before and after each frequency point as the smoothed value

$$\bar{P}_{s_i}(f) = \frac{1}{2U+1} \sum_{u=-U}^U P_{s_i}(f+u). \quad (19)$$

This smoothing filtering can reduce random noise while maintaining a sharp signal [26]. Then, we select a clipping level $\alpha(V_{\max} + V_{\min})$ on the basis of the maximum value $V_{\max}$ and minimum value $V_{\min}$ of $\bar{P}_{s_i}(f)$. In the part greater than the clipping level, the center frequency $f_{i,0}$ is selected as the center position of the subband where the i th signal spectrum is located. Then, combined with the spectral support, the carrier of the i th signal can be determined by

$$\hat{f}_i = f_{i,0} + (S_{i,1} - 0.5)f_p. \quad (20)$$

Algorithm 1 Spectrum Measurement and Parameters' Estimation Algorithm for ACMWC

Input: Number of sources M ; measurement $\mathbf{x}[k]$.

Output: Spectral supports $\mathbf{S}$; signal spectrum $\{\hat{S}_i(f)\}_{i=1}^M$; carriers $\{f_i\}_{i=1}^M$; DOAs $\{\theta_i\}_{i=1}^M$.

- 1: Construct the sensing matrix $\mathbf{G}$ by (11).
- 2: Perform CTF block, initialization: residual $\mathbf{r}^0 = \mathbf{v}$, index set $\mathbf{Z}_0 = \emptyset$.
- 3: **for** $i = 1, \dots, M$ **do**
- 4: $z_i \leftarrow \arg \max_z \|\mathbf{G}^H \mathbf{r}^i\|_1$.
- 5: Check adjacent indexes $z'_i \leftarrow \arg \max_{z \in \mathbf{Z}'} \|\mathbf{G}_{\mathbf{Z}'}^H \mathbf{r}^i\|_1$, $\mathbf{Z}' = [z_i - 1, z_i + 1]$.
- 6: Update the supports $\mathbf{Z}_{i+1} \leftarrow \mathbf{Z}_i \cup \{z_i, z'_i\}$.
- 7: Update residual $\mathbf{r}^{i+1} \leftarrow \mathbf{v} - \mathbf{G}_{\mathbf{Z}_{i+1}} \mathbf{G}_{\mathbf{Z}_{i+1}}^{\dagger} \mathbf{v}$.
- 8: $i \leftarrow i + 1$
- 9: Compute the spectral supports $\mathbf{S}$ by (14).
- 10: Compute the electronic angles $\boldsymbol{\tau}$ by (15) and (16).
- 11: Reconstruct the spectrum $\{\hat{S}_i(f)\}_{i=1}^M$, $f \in [-f_p/2, +3f_p/2]$ by (18).
- 12: Compute the smoothed power spectrum $\bar{P}_{s_i}(f)$ from (19), and search for the center frequency $f_{i,0}$.
- 13: Estimate carrier $\{f_i\}_{i=1}^M$ and $\{\theta_i\}_{i=1}^M$ by (20) and (21), respectively.
- 14: **return** $\mathbf{S}$, $\{\hat{S}_i(f)\}_{i=1}^M$, $\{f_i\}_{i=1}^M$ and $\{\theta_i\}_{i=1}^M$

Then the DOA θ_i can be calculated by

$$\hat{\theta}_i = \arcsin\left(\frac{\hat{\tau}_i}{\hat{f}_i}\right). \quad (21)$$

Algorithm 1 details the process of the proposed algorithm.

It is worth mentioning that the proposed ACMWC has an advantage over other array sub-Nyquist sampling methods, that is, it has good expansibility, which can reduce the number of required elements through channel expansion technology [11].

D. Spatial Separate Scheme

We also give a spatially separated scheme, to reduce the computational complexity. It can be seen that in step 4 of Algorithm 1, the eigenvalue of sensing matrix $\mathbf{G}$ has to be calculated, resulting in a large computational complexity of $L_0^2 P^2 N$. To reduce the computational complexity, we propose a spatially separated SOMP scheme, which calculates l_1 -norm

$$\underbrace{\begin{bmatrix} X_1(e^{j2\pi f T_s}) \\ X_2(e^{j2\pi f T_s}) \\ \vdots \\ X_N(e^{j2\pi f T_s}) \end{bmatrix}}_{\mathbf{X}(e^{j2\pi f T_s})} = \underbrace{\left(\underbrace{\begin{bmatrix} c_{1,0} & \cdots & c_{1,L_0} \\ \vdots & \ddots & \vdots \\ c_{N,0} & \cdots & c_{N,L_0} \end{bmatrix}}_{\mathbf{C}^T} \odot \underbrace{\begin{bmatrix} e^{j2\pi \frac{d_1}{c} \delta} & \cdots & e^{j2\pi \frac{d_1}{c} \delta P} \\ \vdots & \ddots & \vdots \\ e^{j2\pi \frac{d_N}{c} \delta} & \cdots & e^{j2\pi \frac{d_N}{c} \delta P} \end{bmatrix}}_{\mathbf{A}^T} \right)^T}_{\mathbf{G}} \underbrace{\begin{bmatrix} \tilde{S}_{1,0}(f) \\ \vdots \\ \tilde{S}_{P,0}(f) \\ \vdots \\ \tilde{S}_{1,L_0}(f) \\ \vdots \\ \tilde{S}_{P,L_0}(f) \end{bmatrix}}_{\tilde{\mathbf{S}}(f)} \quad (11)$$

Algorithm 2 Spatial Separate SOMP Scheme

```

1: for  $i = 1, \dots, M$  do
2:   for  $p = 1, \dots, P$  do
3:      $z_{i,p} \leftarrow \arg \max_z \|G^H(:, (p-1)L_0 : pL_0)r^i\|_1$ .
4:     Check adjacent indexes
5:      $z'_{i,p} \leftarrow \arg \max_{z \in Z'_p} \|G^H_{Z'_p} r^i\|_1$ ,  $Z'_p = [z_{i,p} - 1, z_{i,p} + 1]$ .
6:     Storage the supports  $Z'(p, :) = \{z_{i,p}, z'_{i,p}\}$ .
7:      $p \leftarrow p + 1$ 
8:    $supp\{z_i, z'_i\} = \arg \max \|Z'\|_1$ 
9:   Update the supports  $Z_{i+1} \leftarrow Z_i \cup \{z_i, z'_i\}$ .
10:  Update residual  $r^{i+1} \leftarrow v - G_{Z_{i+1}} G^\dagger_{Z_{i+1}} v$ .
11:   $i \leftarrow i + 1$ 

```

separately for each point of the grid of electronic angles, and then takes the index of the largest value in P grids as the support set. In this way, the sensing matrix G is divided into $P N \times L_0$ matrices, and the computational cost is reduced to $L_0^2 P N M$, which is reduced by P/M times. The details of the spatially separated SOMP scheme are given in Algorithm 2, which can replace steps 3–8 in Algorithm 1.

E. Computational Complexity

Then, we analyze the computational complexity of the proposed ACMWC and the studied methods. Assuming that the total number of grids $\{\delta p\}_{p=1}^P$ of ACMWC is P , and integer $L = 2L_0 + 1$.

For the proposed ACMWC system, running CTF block costs $N^2(K + N + M)$ multiplications. The SOMP steps in Algorithm 1 cost $L_0^2 P^2 N + L_0 P M^2 N + 10M^2 N + 9M^3 + 2M N K + 3M N^2$ multiplications. And the signal reconstruction costs $2ML^2 + M^2 N + M^3 + 2M N K$ multiplications. So the whole complexity of the proposed Algorithm 1 is $O(L_0^2 P^2 N + L_0 P M^2 N + 11M^2 N + 10M^3 + N^2 K + N^3 + 4M N^2 + 2ML^2 + 2M N K)$.

And the whole complexity of the simplified spatial separate joint estimation algorithm is $O(M P (L_0^2 N + L_0 N M + 2M N) + N^2 K + N^3 + 2M^2 N + 2M N^2 + 2M^3 + 2ML^2 + 2M N K)$. The computational cost is reduced by P/M times compared with Algorithm 1.

While the complexity of the ESPRIT method [20] is $O(N^2 K + N^3 + 8M^2 N + 10M^3 + 2ML^2 + M N (K + 1))$. And the complexity of the CS method [20] is $O(N L_g^2 + N^2 L_g^2 (M + 1) + 2M^2 N^2 + N^4 (M + 1) + 2M^3 + 2ML^2 + 2M^2 N + M N K)$.

Suppose the iteration time of the alternating least-squares (ALS) algorithm is C_p . The complexity of the PARAFAC method [21] is $O(C_p (4M N + 18N^2 M + 16N^3 + M^2 N^2 + M^3) + N^2 K + 2ML^2 + M N K)$. And for the PASS-FD method developed in [16], the whole complexity is $O(C_p (16K^2 N^2 M + 16K^3 N^3 + 4KM^2 N + 2N^4 M + N^6 + 2KM N^2))$.

The complexity of the proposed algorithm is much lower than that of the PASS-FD method [16], but slightly higher than the other three methods. However, it can be found from the experiment in Section IV that the ACMWC system has a better performance in estimation accuracy and signal reconstruction.

TABLE I
MAIN PROPERTIES OF METHODS STUDIED

Methods	ACMWC	ESPRIT [20], CS [20], PARAFAC [21]	PASS-FD [16]	MWC [11]
Array structure	LA MWC	L-shaped MWC	ULA	MWC
Minimal Number of sensors	$2M$	$2M + 1$	$M + 1$	$2M$ channels
Minimal total sampling rate	$2MB$	$(2M + 1)B$	$(M + 1)B$	$2MB$
Reconstruct	DOA, Carrier, Support, Signal	DOA, Carrier, Signal	DOA, Carrier, Power Spectrum	Support, Signal
Source ambiguity	✓	×	✓	None
Coherent Source	✓	×	×	×

F. Comparison With the Studied Method

From Theorem 1, it can be concluded that the minimum number of array elements of ACMWC is $2M$, and the minimum sampling rate of each sensor is $f_s = B$. Therefore, the minimum total sampling rate required for reconstruction is $2MB$. Table I summarizes the main characteristics of ACMWC and comparison methods. The PASS-FD [16] method reconstructs the signal power spectrum without phase information, so the sampling rate is half of other methods. Compared with the traditional MWC system [11], we realize DOA estimation in ACMWC under the same hardware cost. Compared with other array sub-Nyquist sampling methods, the minimum number of elements and the minimum sampling rate are comparable. In addition, the ACMWC system does not require the array elements to be uniformly arranged, but only needs to know the position of the LA elements, which is easy to achieve. And the proposed ACMWC system can still obtain correct parameter estimation results in the presence of source ambiguity and coherent signals, which is more suitable for practical applications.

These methods have a single-channel minimal sampling rate of B , which preserves all the information about signal modulation into the baseband. This is theoretically enough information to perfectly estimate the parameters and reconstruct the signal. For the proposed method and the four compared array sub-Nyquist sampling methods [16], [20], [21], a sampling rate higher than B introduces redundant frequency information. The redundant frequency information outside the baseband will be demodulated into the reconstructed signal during the reconstruction process, which in turn will reduce the carrier frequency estimation and the signal reconstruction effect. Therefore, the effect of sampling rate is not discussed in this work, and the default sampling rate is the best value obtained by performing calculations based on the maximum bandwidth B of the signal.

IV. SIMULATION EXPERIMENTS

In this section, we present numerical simulations demonstrating the feasibility of the proposed ACMWC system. Four series simulation experiments were conducted to investigate the effects of signal-to-noise ratio (SNR), the number of sensors N , and the number of snapshots K on estimation performance, and the estimation capability of the method for coherent signals and ambiguous sources was tested. We compared our method with the ESPRIT and CS methods [20], the PARAFAC method [21], and the PASS-FD method [16]. The receiver of the ESPRIT, CS, and PARAFAC methods is uniform L-shaped array MWC. And the PASS-FD is based on uniform phased delay ULA. We set the total number of sensors to be the same for these methods for a fair comparison. Since Algorithm 2 is the version of Algorithm 1 with reduced computational complexity, we use Algorithm 2 for verification in the following experiments.

Consider $M = 3$ signals with maximum bandwidth $B = 40$ MHz and Nyquist rate $f_{\text{nyq}} = 10$ GHz. The LA consists of $N = 15$ sensors. The pseudorandom sequence $p_n(t)$ alternates at ± 1 with period length $L = 165$ and $f_p = 60$ MHz. The cutoff frequency of LPF is 30 MHz, the number of snapshots $K = 64$ per channel, and the sampling rate is set to $f_s = f_p$. The noise was generated following a Gaussian distribution controlled by SNR. The performance of signal reconstruction is measured by the mean square error (MSE) between the original and reconstructed signals. The estimation performance is accessed by normalized MSE (NMSE) [31] defined as

$$\text{NMSE}_\alpha[\text{dB}] = 10 \log_{10} \left(\frac{1}{J} \sum_{j=1}^J \sum_{i=1}^M \left(\frac{\hat{\alpha}_i^j - \alpha_i}{\alpha_i} \right)^2 \right) \quad (22)$$

where J is the times of Monte Carlo simulations, and $\hat{\alpha}_i^j$ denotes the estimation value of the i th signal carrier or DOA α_i for the j th test.

A. Experiment 1: Influence of SNR

Fig. 2 illustrates the NMSE curves of parameters' estimation, the MSE of reconstruction, and the variance of the estimation varying SNR from -20 to 20 dB, respectively. The Monte Carlo simulation times are set to $J = 500$. The PASS-FD method [16] reconstructs the power spectrum, so the signal reconstruction experiment is not compared with it. The effects of carrier frequency estimation and signal reconstruction of the proposed ACMWC are significantly better than other four array sub-Nyquist sampling methods, and the effect of DOA estimation is slightly better than other methods. As the SNR increases, the estimating error becomes smaller and estimates become more consistent. From the average variances of the estimated carrier frequency and DOA, ACMWC was shown to be less erroneous and more stable. This is because we use both the different mixing information and spatial phase of each channel to estimate parameters, instead of relying solely on spatial phase like other methods, and the estimation results are more accurate and reliable. And the different mixing sequences enhance the randomness of the sensing matrix, which can enhance the robustness of the system. However,

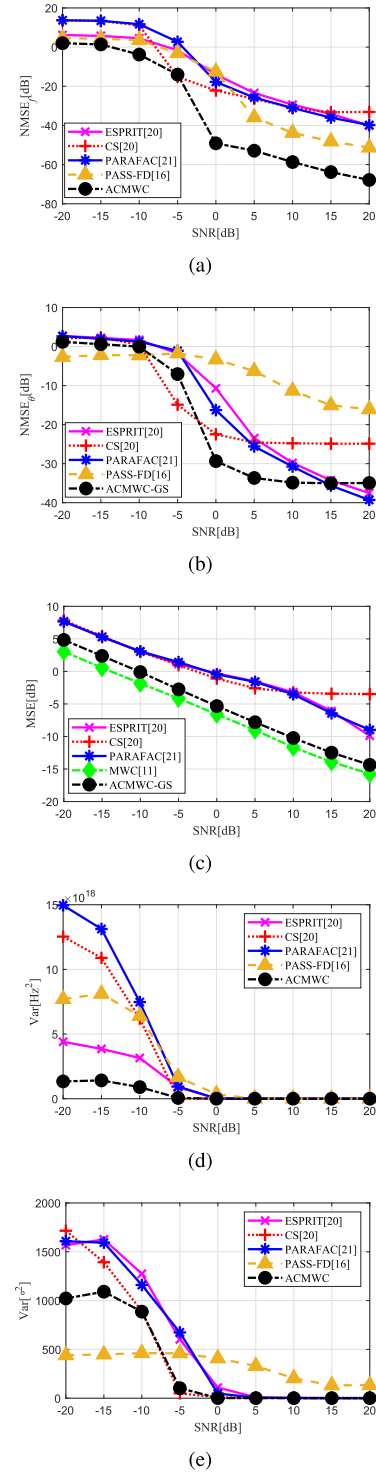


Fig. 2. Influence of SNR on estimation performance. (a) NMSE_f versus SNR. (b) NMSE_θ versus SNR. (c) MSE versus SNR. (d) Average variance of $\hat{f}_i$ versus SNR. (e) Average variance of $\hat{\theta}_i$ versus SNR.

the estimation of DOA is not as accurate as carrier frequency, because the estimation of electronic angles depends on the grid size, and a certain grid errors will occur when the target signals are not on the grid. At the same time, the error of the grid will lead to the deviation of the sensing matrix from the theoretical value, so the performance of signal reconstruction is slightly worse than that of the traditional MWC structure.

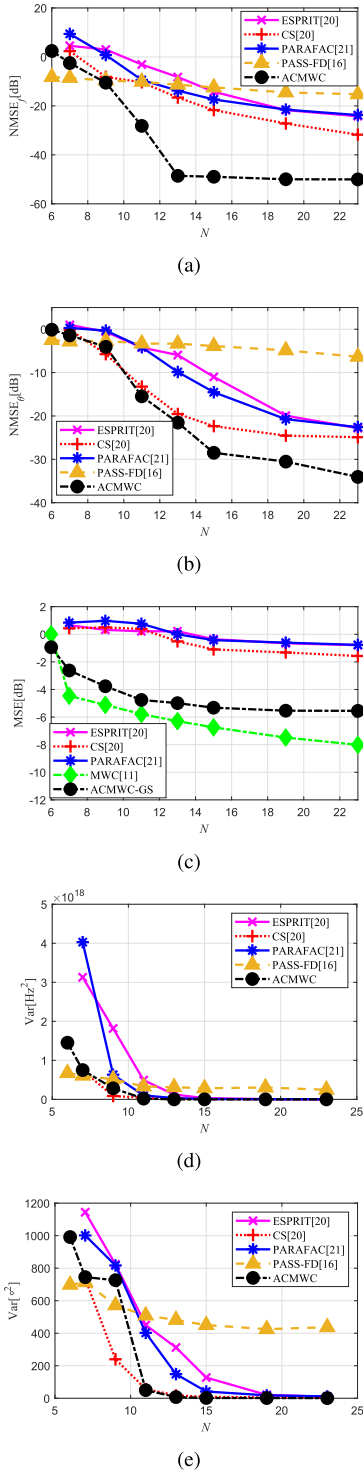


Fig. 3. Influence of N on estimation performance. (a) NMSE_f versus N . (b) NMSE_θ versus N . (c) MSE versus N . (d) Average variance of $\hat{f}_i$ versus N . (e) Average variance of $\hat{\theta}_i$ versus N .

B. Experiment 2: Influence of N

Fig. 3 depicts the NMSE of parameters' estimation, the MSE of reconstruction, and estimation variance varying the number of sensors N per channel from 6 to 23, respectively. The estimation performance improves with the increase in N , and the robustness against noise is enhanced. Among them, the minimum N of ESPRIT, CS [20], and PARAFAC [21] methods is $N = 2M + 1 = 7$. It can be seen that under the

same number of sensors, the proposed ACMWC system has better carrier estimation and signal reconstruction performance and estimation stability than other array sub-Nyquist sampling methods, and the signal reconstruction performance is slightly worse than MWC. In low SNR scenarios, the ACMWC system can get better results when $N \geq 4M$, and the subsequent increase in the number of elements has little effect on estimation accuracy.

C. Experiment 3: Influence of K

Fig. 4 depicts the estimation performance of carries, DOAs, signal reconstruction, and estimation variance when the number of snapshots varies from 16 to 256, respectively. The performance of all the compared methods improves with the increase in K . When K increases from 16 to 64, the parameter estimation and signal reconstruction effects of the ACMWC system are significantly improved. Under the same number of snapshots, the carrier frequency estimation and signal reconstruction performance of the proposed ACMWC system are outperformance than other array sub-Nyquist sampling methods.

D. Experiment 4: Estimation Under Source Ambiguity and Coherent Sources

Then, we test the estimation ability of five array sub-Nyquist sampling methods in the presence of source ambiguity and coherent sources. Consider $M = 5$ mixed coherent and uncorrelated signals with electronic angles $\tau = [-0.08 \ 0.10 \ -0.08 \ -0.08 \ 0.30]e^{10}$. The carrier frequencies of these five sources are given, respectively, by $f_1 = 1.6$ GHz, $f_2 = 1.6$ GHz, $f_3 = 3.2$ GHz, $f_4 = 1$ GHz, and $f_5 = 3.6$ GHz. The corresponding DOAs are set to $\theta_1 = -30^\circ$, $\theta_2 = 40^\circ$, $\theta_3 = -14.5^\circ$, $\theta_4 = -53.1^\circ$, and $\theta_5 = 55^\circ$. Signals 1 and 2 are coherent. The steering vectors of signals 1, 3, and 4 are linearly correlated, and signal 5 is independent of other signals. Fig. 5 gives the true and estimated parameters among the five signals. It can be found that the proposed method can distinguish coherent signals 1 and 2, while the other two methods fail to identify the coherent signals. Due to the rank deficiency of the array manifold matrix caused by the correlation of steering vectors, the ESPRIT-, PARAFAC-, and CS-based methods cannot estimate parameters correctly in the case of source ambiguity between signals 1, 3, and 4. However, both the ACMWC and PASS-FD methods can distinguish the ambiguous sources, and obviously the performance of ACMWC to estimate the parameters is better.

E. Experiment 5: Different Signal Models

The proposed method can be applied to narrowband signals of any signal form that satisfies the band limited to $\mathcal{B}$ as defined in Section II, and perfect parameter estimation and signal reconstruction can be achieved when Theorem 1 is satisfied. And this conclusion also applies to signal forms that can be approximated as band-limited in the frequency domain, such as binary phase shift keying (BPSK) signals and linear frequency modulation (LFM) signals. These signals are commonly used in communications and radar, and most of

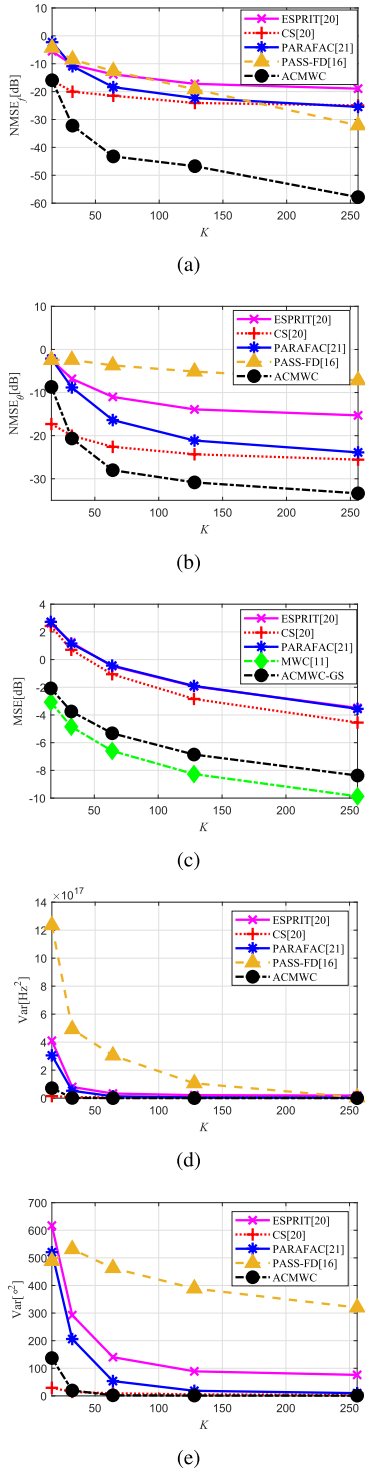


Fig. 4. Influence of K on estimation performance. (a) NMSE_f versus K . (b) NMSE_θ versus K . (c) MSE versus K . (d) Average variance of $\hat{f}_i$ versus K . (e) Average variance of $\hat{\theta}_i$ versus K .

their energy is concentrated in the main waveband. Finally, we tested the estimation of the ACMWC system for different signal models. In addition to the Sinc signal used in the previous tests, the BPSK signal and the LFM signal were also used as test signals. The performance of the carrier frequency, DOA estimation, and signal reconstruction varying SNR from -20 to 20 dB for each of the three test signals is presented in Fig. 6. It can be seen that the Sinc signal

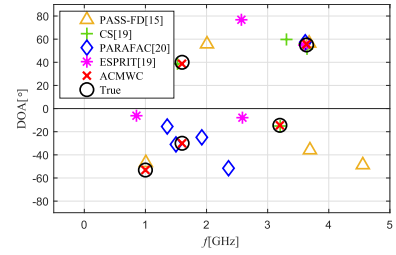


Fig. 5. True and estimated carrier frequencies and DOAs in the presence of source ambiguity and coherent sources, with $N = 25$, $K = 64$, and $\text{SNR} = 10$ dB.

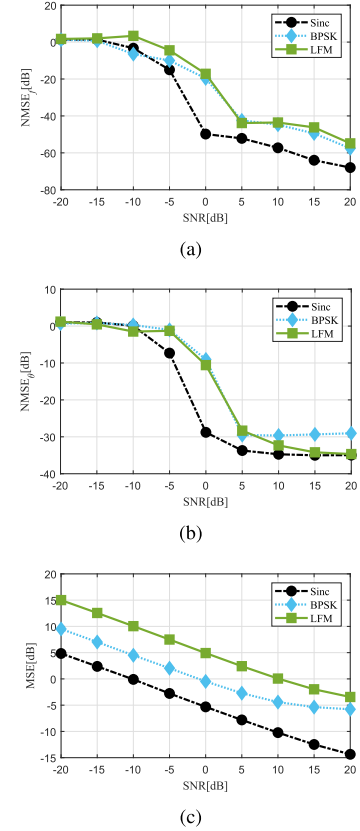


Fig. 6. Influence of different signal models. (a) NMSE_f of different signal models versus SNRs. (b) NMSE_θ of different signal models versus SNRs. (c) MSE of different signal models versus SNRs.

gives the best estimate. This is because the Sinc signal is an ideal band-limited signal in the frequency domain, which satisfies our definition of the signal model in Section II. The BPSK and LFM signals are approximately band-limited in the frequency domain. Although most of their energy is in the main lobe, the frequency components outside the main lobe cannot be reconstructed by the ACMWC system, and the excess frequency components are also mixed into the main lobe, which affects the signal reconstruction performance. Therefore, the performance of parameter estimation and signal reconstruction for BPSK and LFM signals is inferior to that of the Sinc signal.

V. HARDWARE EXPERIMENTS

We then implemented a hardware platform to test the effectiveness of the principle. We first describe the hardware environment, and then analyze the nonideal characteristics of the physical devices that may affect the results. In the

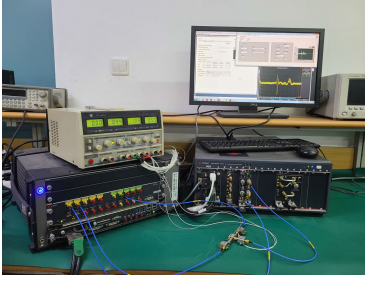


Fig. 7. Photograph of sampling hardware.

hardware experiments, we conducted six series of experiments to investigate the effects of SNR, the number of sensors N , and the number of snapshots K on the estimation performance, and the estimation results were tested for different DOA and carrier frequency parameters, different signal models, and the presence of coherent and ambiguous sources.

A. Hardware Environment Description

First, we give a brief introduction to our hardware platform. Since the main concern of this work is the sub-Nyquist sampling and signal processing at the back-end rather than the working process of the front-end receiver, we elide the down-conversion at the front-end when building the hardware verification platform and use the waveform generator to generate the IQ signals directly, such as in the hardware experiment in [32]. The platform included an arbitrary waveform generator Keysight M8190A controlled by an M9037A PXIe embedded controller, an analog pretreatment circuit, and a Keysight M9243A PXIe modular oscilloscope. The analog pretreatment circuit consisted of two ZX60-V62+ amplifier, a ZX05-42MH-S+ RF mixer, and a 50-MHz ZX75LP-50-S+ LPF. The ZX60-V62+ RF amplifier operates at frequencies from 50 MHz up to 6 GHz with relatively flat gain. The ZX05-42MH-S+ supports signals in the spectral range of 5–4200 MHz, with low conversion loss and high L-R isolation. The photograph of the sampling hardware is shown in Fig. 7.

The block diagram of the proposed sampling system is shown in Fig. 8. Because M8190A has only two generation channels, and the M8190A has only two generation channels, one channel is used to generate the test signal and the other channel is used to generate the mixing sequence. Therefore, we can only perform signal generation, analog pretreatment, and sub-Nyquist sampling for one physical channel over a period of time. So we cyclically generate the received signals from multiple arrays through M8190A, using one physical channel for analog pretreatment. Although the signals are not generated simultaneously, this does not affect the validation of the effectiveness of the principle [33].

First, the IQ signals and the mixing sequence are generated by M8190A. The trigger signal was generated by M8190A along with the mixing sequence and was input to the external trigger interface of the oscilloscope. The ZX60-V62+ amplifiers are then used to amplify the signal and the mixing sequence and were powered by an Agilent U8001A dc power supply. After being amplified, the signal is mixing with mixing sequence $p(t)$. The repetition frequency of $p(t)$ is

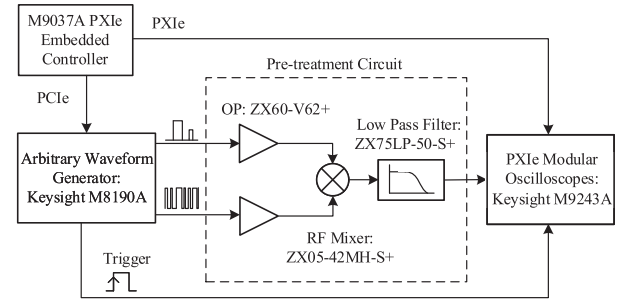


Fig. 8. Block diagram of the proposed sampling system.

$f_p = 60$ MHz with a length of $L = 165$ bits. Then the mixed signal is filtered by a 50-MHz cutoff frequency LPF ZX75LP-50-S+. The filtered signal is sampled by a Keysight M9243A oscilloscope with a sampling rate of $f_s = 150$ MHz. Finally, we used the samples to estimate parameters and reconstruct signals on the computer.

B. Nonideal Characteristics' Analysis

In practice, nonideal characteristics of RF devices can affect the performance of parameter estimation and signal reconstruction, resulting in hardware experiments yielding less effective results than simulation tests under ideal conditions. The main nonideal influences in the ACMWC system include the voltage standing wave ratio (VSWR) of the amplifier and mixer, the LO-RF and LO-IF isolation of mixer, the nonideal amplitude and phase frequency characteristics of filter, and the introduced noise in the sampling process. We then briefly analyze these nonideal factors as follows.

The frequency spectrum of the input signal $x(t)$ is defined as $X(f)$. And the frequency spectrum of the mixing sequence $p(t)$ is defined as $P(f)$. First, both the input signal and the mixing sequence are amplified by the amplifier and input to the RF port and LO port of the mixer, respectively. The VSWR can measure the degree of signal reflection caused by impedance mismatch at the interface. Due to the VSWR of the amplifier and the mixer, the resulting input signal and mixing sequence into the mixer can be represented as follows:

$$\begin{aligned} X_{MX}(f) &= v_{AM}(f)v_{RF}(f)X(f) \\ P_{MX}(f) &= v_{AM}(f)v_{LO}(f)P(f) \end{aligned} \quad (23)$$

where $v_{AM}(f)$, $v_{RF}(f)$, and $v_{LO}(f)$ denote the attenuation factors of the amplifier, the RF port of the mixer, and the LO port of the mixer, respectively.

The mixer's LO-RF isolation indicates the level at which the signal from the LO port feeds through to the RF port. If there is an impedance mismatch at the RF port, the feedthrough signal will be reflected and superimposed on the input signal, resulting in a redundant frequency component. As the signal energy gained by both the feedthrough and reflection processes is very small, its effect can be neglected.

The signal after mixing can be expressed as

$$Z_{MX}(f) = d_c(f)[X_{MX}(f) * P_{MX}(f)] \quad (24)$$

where $d_c(f)$ denotes the conversion loss of the mixer.

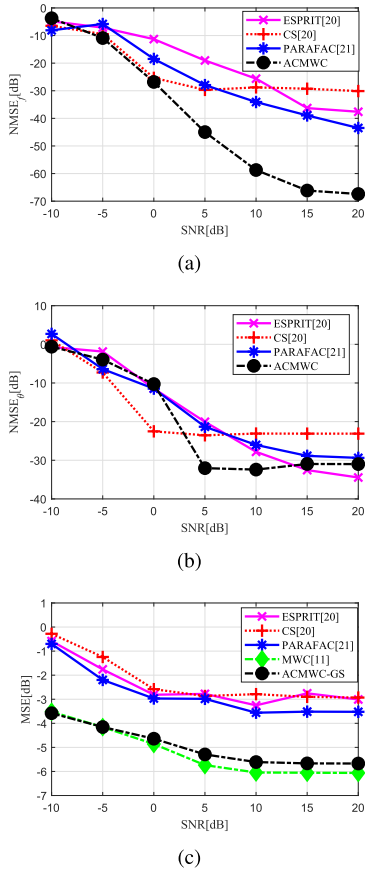


Fig. 9. Influence of SNR on hardware experiment estimation performance. (a) NMSE_f versus SNRs. (b) NMSE_θ versus SNRs. (c) MSE versus SNRs.

LO-IF isolation indicates the level of LO signal feedthrough to the IF output port. The IF output port of the mixer will have a portion of the signal from the mixer function. The IF output port of the mixer will therefore have a portion of the signal from the mixing sequence. And we also consider the white noise signal introduced during the sampling process. The final output of the mixer can be expressed as

$$\bar{Z}_{MX}(f) = Z_{MX}(f) + w_{LI}(f)P_{MX}(f) + N(f) \quad (25)$$

where $w_{LI}(f)$ is the degree of feedthrough from the LO port to the IF output port, and $N(f)$ is the noise introduced.

Finally, after low-pass filtering, the signal can be expressed as

$$\begin{aligned} Y(f) &= \bar{Z}_{MX}(f)H(f) \\ &= d_c(f)[X_{MX}(f) * P_{MX}(f)]H(f) \\ &\quad + w_{LI}(f)P_{MX}(f)H(f) + N(f)H(f). \end{aligned} \quad (26)$$

where $H(f)$ is the frequency response of the LPF.

It should be noted that the nonideal amplitude and phase frequency response of the filter has the most serious impact on the ACMWC system and can lead to estimation and reconstruction failures. Therefore, we compensate for the filter's nonideality in the frequency domain using the calibration method in [34], specifically by digitally compensating the sampled values according to the amplitude and phase frequency response of the filter.

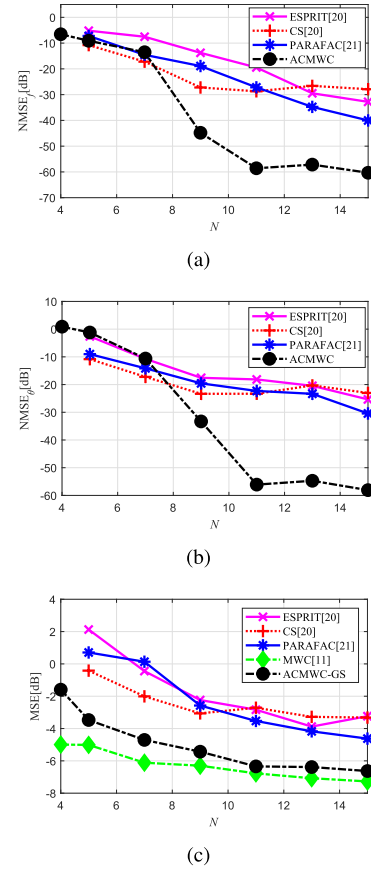


Fig. 10. Influence of N on hardware experiment estimation performance. (a) NMSE_f versus N . (b) NMSE_θ versus N . (c) MSE versus N .

C. Hardware Experiment 1: Influence of SNR

Suppose there are $M = 2$ signals with carrier $f_1 = 1.6$ GHz, $f_2 = 3.5$ GHz, the bandwidth of these two signals is $B_1 = 40$ MHz, $B_2 = 20$ MHz, and DOA $\theta_1 = -30^\circ$, $\theta_2 = 56^\circ$. Set the total number of sensors $N = 9$, the number of snapshots $K = 64$, and the sampling rate $f_s = 150$ MHz. We compensate for the filter's nonideality in the frequency domain using the calibration method in [34].

Fig. 9 shows the performance of the carrier frequency, DOA estimation, and signal reconstruction, respectively. As can be seen, the estimation and performance effect of these methods improves as the SNR increases. Similar to the simulation results, the proposed ACMWC performs better in carrier frequency estimation and signal reconstruction than other array sub-Nyquist sampling methods.

D. Hardware Experiment 2: Influence of N

We then verified the variation pattern of the estimation performance with the number of sensors N in hardware experiments. The hardware experiments were performed ten times versus N from 4 to 15. The curve trends of the hardware results can be seen in Fig. 10, which is consistent with the simulation results, further verifying the validity of the proposed system.

E. Hardware Experiment 3: Influence of K

Similarly, we verified the pattern of estimation performance with the number of snapshots K on a hardware platform.

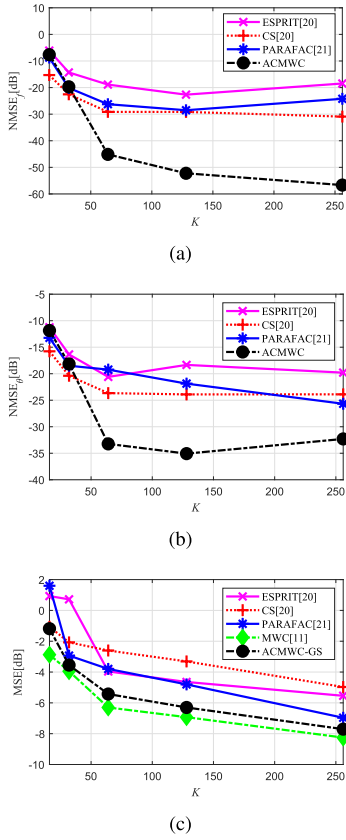


Fig. 11. Influence of K on hardware experiment estimation performance. (a) $NMSE_f$ versus K . (b) $NMSE_\theta$ versus K . (c) MSE versus K .

The hardware experiments was performed ten times versus K from 16 to 256. Fig. 11 shows the performance of the carrier frequency, DOA estimation, and signal reconstruction, respectively. It can be seen that the estimation results are getting better with an increasing K . In particular, the ACMWC method shows a significant improvement in parameter estimation and signal reconstruction as K increases for $K < 64$. At $K \leq 64$, the improvement of estimation performance becomes moderate and becomes almost flat. For $K > 32$, the proposed ACMWC method has significantly better estimation and reconstruction results than the other methods.

F. Hardware Experiment 4: Estimation Under Source Ambiguity and Coherent Sources

Then, we verify the ability of the proposed method to estimate ambiguous sources and coherent sources on hardware platform with $SNR = 10$ dB. The experimental tests were performed five times. The carrier frequencies of these five sources are given, respectively, by $f_1 = 1.6$ GHz, $f_2 = 1.6$ GHz, $f_3 = 3.2$ GHz, $f_4 = 1$ GHz, and $f_5 = 3.6$ GHz. The corresponding DOAs are set to $\theta_1 = -30^\circ$, $\theta_2 = 40^\circ$, $\theta_3 = -14.5^\circ$, $\theta_4 = -53.1^\circ$, and $\theta_5 = 55^\circ$. Signals 1 and 2 are coherent, and signals 1, 3, and 4 have linearly correlated steering vectors. Fig. 12 gives the true and estimated carriers and DOAs.

G. Hardware Experiment 5: Different DOAs and Carriers

To verify the effectiveness of the proposed method for other DOA and carrier frequency settings, ten groups of signals

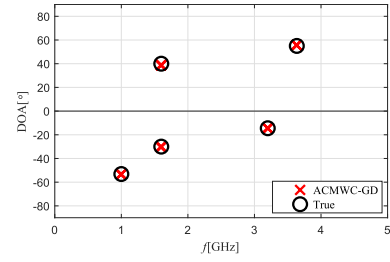


Fig. 12. True and estimated carrier frequencies and DOAs of hardware experiment, with $N = 25$, $K = 64$, and $SNR = 10$ dB. Carrier frequencies were set to [1.6 1.6 3.2 1 3.6] GHz and DOAs to $[-30^\circ 40^\circ -14.5^\circ -53.1^\circ 55^\circ]$.

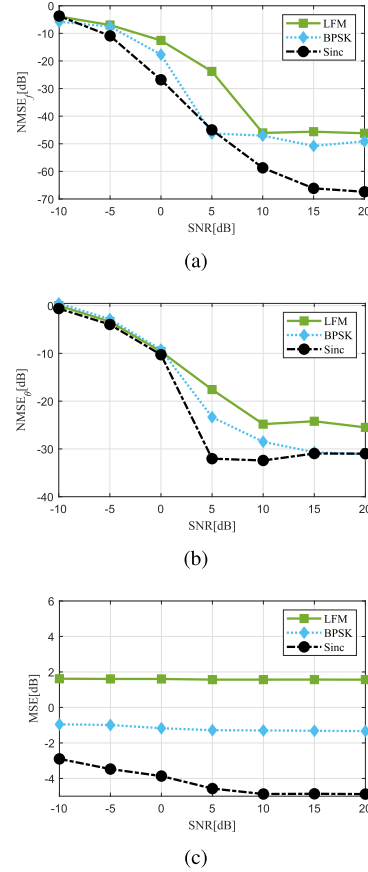


Fig. 13. Influence of different signal models on hardware experiment estimation performance. (a) $NMSE_f$ of different signal models versus SNRs. (b) $NMSE_\theta$ of different signal models versus SNRs. (c) MSE of different signal models versus SNRs.

with different carrier frequency and DOA parameters were selected for experimental verification. The $SNR = 10$ dB was set, and the real parameters and estimated results are shown in Table II. According to the experimental results, the built hardware ACMWC system can correctly estimate the DOA and carrier frequency parameters for different parameter settings.

H. Hardware Experiment 6: Different Signal Models

Finally, we tested the estimation of the ACMWC system for different signal models in hardware experiment. In addition to the Sinc signal used in the previous tests, the BPSK signal and the LFM signal were also used as test signals. The experimental parameters were set as in Hardware Experiment 1.

TABLE II
REAL PARAMETERS AND ESTIMATED RESULTS

No.	True θ_i (°)	Estimated $\hat{\theta}_i$ (°)	True f_i (GHz)	Estimated $\hat{f}_i$ (GHz)
1	[35 - 40]	[36.870 -43.441]	[1 3.2]	[1.000 3.200]
2	[60 20]	[58.647 18.833]	[2 3.4]	[1.991 3.408]
3	[-60 45]	[-54.593 39.822]	[0.6 2.5]	[0.613 2.498]
4	[-15 50]	[-21.940 48.656]	[0.8 2.8]	[0.803 2.797]
5	[-45 70]	[-46.320 71.826]	[1.4 2]	[1.383 2.000]
6	[40 - 45]	[-46.706 39.410]	[1.1 2.2]	[1.103 2.198]
7	[38 - 25]	[37.733 -22.783]	[1.8 3.1]	[1.797 3.099]
8	[58 - 35]	[61.683 -33.738]	[1.7 2.7]	[1.704 2.701]
9	[35 - 58]	[32.653 -57.384]	[1.9 3.15]	[1.900 3.151]
10	[25 66]	[23.293 63.890]	[1.5 2.9]	[1.517 2.896]

The performance of the carrier frequency, DOA estimation, and signal reconstruction varying SNR from -10 to 20 dB for each of the three test signals is presented in Fig. 13. Similar to the simulation results, the experimental results show that the Sinc signal is the best estimate.

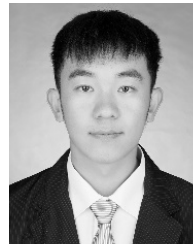
VI. CONCLUSION

In this article, an ACMWC system is proposed for spectrum measurement and DOA estimation. By setting different mixing sequences in each receiving channel of the LA receiving structure, the signal spectral supports and electron angles can be estimated simultaneously. Through the information in supports and the electronic angles, the source ambiguity and coherent signals problem in the array parameter estimation can be solved. Compared with other array sub-Nyquist sampling methods, ACMWC relaxes the restrictions on the signal model and array and has a wider range of applications. In addition, a spatially separated SOMP scheme is proposed to reduce the computational complexity. The simulation results show that compared with other methods, ACMWC can significantly improve the performance of carrier frequency and spectrum estimation, and it can distinguish the ambiguous sources and coherent sources. We have also implemented a hardware platform, the results of which verify the feasibility of the proposed method. In the future, we could combine channel extension technology to reduce the number of elements required.

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